Before v2: Three Fatal Defects (0.2% Recall)

1. Global Rescaling Deflation:
$S \leftarrow S/\mathrm{std}(S)$ divided by the sample standard deviation. A single bright star residual inflated $\mathrm{std}(S)$, deflating every other real detection below 5σ.

2. Unconstrained $75''$ Dipole Pairing:
Any negative dip was paired with any positive peak within $75''$. Real fading stars were glued to unrelated noise and mislabelled as fake movers.

3. Fixed $3'$ Bright-Star Mask:
Failed on 1st-magnitude stars. The halo of ϵ Indi A bled past $3'$, spawning 44 false candidate detections.

Decision A: The Calibrated Detector (69.3% Recall)

1. Robust MAD Scaling (`step3_zogy.py`):
Normalized by median absolute deviation: $S/(1.4826 \cdot \mathrm{MAD})$. Resistant to outliers; bright residuals no longer suppress faint transients.

2. Physical Kinematic Pairing (`config.py`):
Separation bounded by realistic stellar motion ($\Delta \leq \mu_{\max} \Delta t$) and flux ratio $\leq 2.0\times$. Fading stars cleanly become negative monopoles.

3. Exponential Dynamic Masks (`catalog_tools.py`):
Mask radius grows exponentially with infrared brightness up to $10'$. Eliminates all 44 false candidates around ϵ Indi A.

Measured Scientific Impact Across 18 DR1 Benchmark Fields:

Planted objects recovered strictly: **surged from 0.2% to 69.3%**

- Brightenings: **96.7%**
- Fadings: **95.3%**
- Movers: **56.0%**